

Daniel McCabe

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EDUCATION

Virginia Tech, College of Engineering

Master of Engineering in Computer Science & Applications – Cybersecurity (Accelerated)

Bachelor of Science in Computer Science, Minors in Cybersecurity and Mathematics

Blacksburg, VA

Expected December 2026

December 2025

GPA: 3.97 – Major Rank: 32 of 906; President's List and Dean's List

Relevant Coursework: Computer Systems, Network Security, Network Architecture, Software and System Security, Database Management Systems, AI Tools for Software Engineering

PROJECTS

Tollgate – Multi-Tenant API Gateway

September 2026

[Live](#) | [GitHub](#)

- Built and deployed a multi-tenant reverse proxy for model API traffic (Python, FastAPI, PostgreSQL, Redis) on AWS ECS Fargate, with every resource declared in Terraform and a GitHub Actions pipeline authenticating by OIDC that migrates, smoke-tests, and automatically rolls back — about five minutes from merge to live
- Cut upstream spend 59–96% with a two-tier request cache serving hits in 14.6ms against 122ms, and enforced per-tenant rate limits and hard spend budgets in integer micro-cents, reconciled to zero error at settlement
- Scored prompt-injection detection across a 1,200-case benchmark against the regex baseline it had to beat, raising recall from 0.374 to 0.666 at higher precision, and instrumented every request with OpenTelemetry

durable-queue – Postgres Job Queue Library

September 2026

[GitHub](#)

- Wrote a Postgres-only job queue in Python with exactly-once semantics for transactional tasks: atomic claiming via FOR UPDATE SKIP LOCKED, crash recovery through leases and heartbeats, and three-layer idempotency
- Raised four-worker throughput 27x through profiling (162 to 4,452 jobs/sec with every commit fsynced) and cut p50 enqueue-to-start latency from 719ms to 6.9ms using LISTEN/NOTIFY; diagnosed a load-only stall to three stacked causes, including an index fix turning a 5.18ms sort into a 0.052ms index-ordered scan

Release Radar – Full-Stack Web Application

October 2025 – Current

[Project Website](#) | [GitHub](#)

- Built and shipped a production watchlist application (React, TypeScript, FastAPI, PostgreSQL) with Firebase authentication, Stripe subscription billing, a social graph, and five async background jobs syncing three external APIs
- Built the release calendar across a user's tracked titles, loading a 150-title library in 31ms instead of 957ms by replacing a request-per-show fan-out with one windowed endpoint (123 HTTP requests to 1, 490 SQL queries to 4)
- Rebuilt the profile page to render in 9ms on 4.8KB rather than 71ms on 190KB, returning preview rows server-side instead of shipping the user's full watchlist and watched list

MCCS Sponsored Graduate Capstone – Team Lead

August 2026 – Current

- Leading a graduate capstone team sponsored by U.S. Marine Corps Community Services through requirements, design, and delivery of an augmented reality application surfacing real-time product information for shoppers

AI Security Research – XSS Vulnerability Analysis

January 2026 – May 2026

- Built a Python pipeline generating 480 code samples across three LLMs, deploying each in an isolated Docker container and attacking it with standard and obfuscated XSS payloads under static and dynamic analysis
- Found all three models produced exploitable XSS at 5.4–6.6% payload hit rates; one security clause added to the prompt eliminated exploitable findings entirely for one model while the others only fell to 3.3–4.3%

WORK EXPERIENCE

Basis Digital Biotech

Boston, MA

Cybersecurity Intern

May 2025 – August 2025

- Automated monthly security threat reports using Python and the Microsoft Graph API, delivering executive-level summaries of Defender alerts, phishing activity, and user risk scores to client stakeholders
- Monitored Microsoft Defender for Endpoint across client environments and helped clients achieve a Microsoft Secure Score 56% above the industry average for similarly sized organizations; ran phishing campaigns yielding ~20% better detection

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, C, SQL, MATLAB

Frameworks and Libraries: React, FastAPI, Express.js, SQLAlchemy, Tailwind CSS, NumPy, Pandas

Cloud and Infrastructure: AWS (ECS Fargate, ALB, IAM), Terraform, Docker, GitHub Actions, Linux, Git

Data and Tools: PostgreSQL, Redis, OpenTelemetry, Grafana, REST APIs, Microsoft Defender, Microsoft Graph API